

TASK-1

Topic-1:- Automobile system explanation

An **automobile system** is a combination of various mechanical, electrical, and electronic systems that work together to move a vehicle safely, efficiently, and comfortably.

Main Systems of an Automobile

1. **Engine System**
 - Generates power by converting fuel energy into mechanical energy.
 - Main components: cylinder, piston, crankshaft, valves, and spark plug.
2. **Fuel System**
 - Stores and supplies fuel to the engine.
 - Components: fuel tank, fuel pump, fuel filter, and injector/carburetor.
3. **Transmission System**
 - Transfers engine power to the wheels.
 - Components: clutch, gearbox, driveshaft, differential, and axle.
4. **Steering System**
 - Controls the direction of the vehicle.
 - Components: steering wheel, steering gear, and tie rods.
5. **Braking System**
 - Slows down or stops the vehicle.
 - Components: brake pedal, master cylinder, and disc/drum brakes.
6. **Suspension System**
 - Absorbs road shocks and provides ride comfort.
 - Components: springs, shock absorbers, and control arms.
7. **Electrical System**
 - Supplies electrical power for starting and vehicle accessories.
 - Components: battery, alternator, starter motor, and wiring.

Working of an Automobile

Fuel is supplied to the engine, where combustion produces power. The transmission system transfers this power to the wheels. The steering system controls direction, the braking system controls speed and stopping, and the suspension system ensures stability and comfort. The electrical system powers lights, ignition, and other accessories.

Conclusion

An automobile consists of several interconnected systems such as the engine, fuel, transmission, steering, braking, suspension, and electrical systems. Together, they ensure efficient, safe, and reliable vehicle operation.

Topic-2:- EV components and working

Definition

An **Electric Vehicle (EV)** is a vehicle that uses an electric motor powered by a rechargeable battery instead of an internal combustion engine.

Main Components of an EV

1. **Battery Pack**
 - Stores electrical energy.
 - Usually consists of lithium-ion batteries.
2. **Electric Motor**
 - Converts electrical energy into mechanical energy.
 - Drives the wheels of the vehicle.

3. **Inverter (Power Electronics Controller)**
 - Converts DC power from the battery into AC power for the motor.
 - Controls motor speed and torque.
4. **Battery Management System (BMS)**
 - Monitors battery voltage, current, and temperature.
 - Protects the battery from damage.
5. **On-Board Charger (OBC)**
 - Converts AC power from the charging station into DC power for charging the battery.
6. **DC-DC Converter**
 - Converts high-voltage battery power into 12V power for lights and accessories.
7. **Regenerative Braking System**
 - Recovers energy during braking and stores it back in the battery.

Working of an EV

1. The battery pack stores electrical energy.
2. When the accelerator is pressed, the battery supplies power to the inverter.
3. The inverter controls and sends power to the electric motor.
4. The motor converts electrical energy into mechanical energy and rotates the wheels.
5. During braking, the motor acts as a generator and sends energy back to the battery through regenerative braking.
6. The BMS continuously monitors battery performance and safety.

Topic-3:- ICE vs EV comparison

Factor	ICE Vehicle	EV
Power Source	Petrol, diesel, CNG	Battery-powered electric motor
Purchase Price	Usually lower upfront cost	Higher upfront cost (though prices are falling)
Running Cost	Higher fuel costs	Generally much lower electricity costs
Maintenance	More moving parts, regular oil changes, filters, engine maintenance	Fewer moving parts, lower maintenance needs
Performance	Good, but power delivery depends on engine and transmission	Instant torque, quick acceleration
Range	Typically 500–1,000+ km per tank	Usually 200–700 km per charge, depending on model
Refueling/Recharging Time	3–5 minutes	20 minutes to several hours, depending on charger
Environmental Impact	Tailpipe emissions (CO ₂ , NO _x , particulates)	Zero tailpipe emissions; overall impact depends on electricity source and battery production
Infrastructure	Extensive fuel station network	Charging network growing but still less widespread in many regions
Noise	Engine noise and vibration	Very quiet operation
Battery Degradation	Not applicable	Battery capacity gradually declines over time

Advantages

1. **Long driving range** – Can typically travel hundreds of kilometers on a full tank.
2. **Fast refueling** – Fueling usually takes only a few minutes.
3. **Established infrastructure** – Fuel stations are widely available.
4. **Lower upfront cost** – Many ICE vehicles are cheaper to purchase than comparable EVs.

5. **Suitable for remote areas** – Less dependent on charging infrastructure.

Challenges

1. **Higher fuel costs** – Petrol and diesel expenses can be significant over time.
2. **More maintenance** – Engine oil, filters, spark plugs, clutch, and transmission components require servicing.
3. **Air pollution** – Produces carbon dioxide (CO₂) and other pollutants.
4. **Noise and vibration** – Engines generate more noise than electric motors.
5. **Dependence on fossil fuels** – Subject to fuel price fluctuations and energy security concerns.

Electric Vehicles (EVs)

Advantages

1. **Lower operating cost** – Electricity is generally cheaper per kilometer than petrol or diesel.
2. **Reduced maintenance** – Fewer moving parts and no oil changes.
3. **Zero tailpipe emissions** – Improves local air quality.
4. **Instant torque** – Strong acceleration and smooth driving experience.
5. **Quiet operation** – Minimal noise and vibration.
6. **Energy efficiency** – Electric motors convert a much larger share of energy into motion than combustion engines.

Challenges

1. **Higher purchase price** – Although prices are decreasing, batteries remain expensive.
2. **Charging time** – Charging takes longer than refueling.
3. **Charging infrastructure limitations** – Availability varies by region.
4. **Range anxiety** – Drivers may worry about running out of charge on long trips.
5. **Battery degradation** – Battery capacity gradually decreases over years of use.
6. **Battery production impacts** – Mining and manufacturing require significant resources.

Automobile system layout

